# 🏛️ House Rules — Complete User Guide

> A beginner-friendly guide: AI manages your company's "house rules" For erpilot v3.25+ Reading time: 15 min

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## 🤔 What are "House Rules"? Why do you need them?

Just like every household has rules (no snacks after 11pm, wash hands at home), **every company has its own "business rules"**:

- PO over NT$100k → boss approves
- WO without Recipe → cannot release to factory floor
- SO discount over 5% → manager reviews
- Customer credit limit exceeded → no new order
- Invoice must be issued before shipping

Traditional approach: **rules live in employees' heads**. Problems:

- 😰 New hires don't know
- 😰 Veterans forget
- 😰 No handoff between roles
- 😰 People sneak around when boss is away
- 😰 Hard to find accountability after incidents

**erpilot's "House Rules" put rules into the system**:

- ✅ Employees follow rules — error rate drops 80%
- ✅ Boss sleeps soundly (system auto-blocks violations)
- ✅ No retraining when staff changes (rules live in system)
- ✅ Manager can emergency-override with audit log
- ✅ ISO / GMP audits → one-click full trail export

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## 🆚 How competitors do it (and why erpilot is different)

| ERP | How to set rules | Drawback |
|---|---|---|
| **SAP B1** | Consultant changes code | Wait 1 month + cost NT$50-200k |
| **鼎新** | Hidden config screens | Rigid conditions; can't handle edge cases |
| **NetSuite** | Write JavaScript (SuiteScript) | Beginners can't do it |
| **Odoo** | Python expressions | Dangerous + requires coding |
| **erpilot ✨** | **👇 Choose any of 3** | **Anyone can edit, instant effect** |

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## 🚀 3 Ways to Set House Rules in erpilot

### Method A: Click in the UI (easiest)

1. Log in
2. Sidebar → "✅ Approvals" (House Rules share this page in v3.25)
3. Switch to "⚙️ Rules" tab
4. Click "➕ Add Rule"
5. Fill the form:

```
┌──────────────────────────────────────┐
│ Name:      No PO over $50k           │
│ Trigger:   [PO Create              ▼] │
│ Condition: [field_compare          ▼] │
│   Field:     amount                  │
│   Operator:  <= (less or equal)      │
│   Value:     50000                   │
│ Action:    [block                  ▼] │
│ Message:   Amount too high, please split │
│ Override:  [manager                ▼] │
│ Active:    ✅                        │
│                                      │
│           [✓ Save]  [✗ Cancel]       │
└──────────────────────────────────────┘
```

6. Save → **Instantly effective** (no restart, no deploy)

### Method B: Talk to AI (natural language — erpilot's signature)

Don't want to fill forms? **Just talk**:

1. Click "💬 AI Assistant" on sidebar
2. Type:

> "Our company should require manager approval when SO discount exceeds 5%"

3. AI responds with ConfirmCard:

```
┌──────────────────────────────────────┐
│ 🤖 I'll set this house rule:         │
│                                      │
│   Name:    SO discount > 5% needs mgr│
│   Trigger: Sales Order create        │
│   Condition: discount_pct > 0.05     │
│   Action:  require_approval          │
│   Override: manager                  │
│                                      │
│   [✓ Confirm] [✗ Cancel]             │
└──────────────────────────────────────┘
```

4. Click [✓ **Confirm**] → Rule goes live
5. Next time a salesperson creates an SO with >5% discount → auto-routed to approval

> 🚀 **This is erpilot's most powerful feature**: describe needs in plain words, AI translates to system rules. No SQL, no JavaScript, no Python required.

## Method C: API / Plugin (for engineers)

Need a custom condition (e.g., credit limit algorithm)? Engineers can plug in new condition types:

```python
from app.services.policy_engine import register_condition

async def check_credit_limit(params, context, db):
    """Check customer credit limit (plugin example)"""
    from app.models.crm_sales import Customer
    from sqlalchemy import select
    cust_id = context.get('customer_id')
    if not cust_id:
        return False
    cust = (await db.execute(
        select(Customer).where(Customer.id == cust_id)
    )).scalar_one_or_none()
    if not cust:
        return False
    return context.get('amount', 0) <= cust.credit_limit

register_condition("credit_check", check_credit_limit)
```

After this, users will see "credit_check" as a condition option in the UI.

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## 🧱 The 4 Building Blocks of a House Rule

### 1️⃣ Trigger

"**What action triggers this rule?**" 16+ built-in triggers:

| Category | Triggers | When |
|---|---|---|
| Production | `wo.release` / `wo.complete` / `wo.cancel` | WO actions |
| Purchase | `po.create` / `po.approve` / `po.receive` / `po.cancel` | PO actions |
| Sales | `so.create` / `so.confirm` / `so.ship` / `so.cancel` | SO actions |
| Inventory | `inventory.delete` / `inventory.transfer` | Delete part / transfer |
| Accounting | `journal.post` / `ar.create` / `ar.collect` | Posting / invoicing / collection |
| CRM | `lead.convert` / `opportunity.stage_changed` | Sprout convert / Chase advance |

### 2️⃣ Condition

"**When does the rule apply?**" 5 built-in types:

| Type | Use | Example |
|---|---|---|
| **always** | Always trigger | "PO must always have remark" |
| **has_bom** | Pass only if Recipe exists | "WO release needs Recipe" |
| **field_compare** | Compare fields | "amount > 100k" |
| **count_check** | Count list length | "PO at least 1 item" |
| **custom** | Plugin-defined | "Credit limit check" |

`field_compare` operators: `gt` / `gte` / `lt` / `lte` / `eq` / `ne`

### 3️⃣ Action

"**What happens on violation?**" 4 types:

| Action | Behavior | Use case |
|---|---|---|
| 🟢 **allow** | Just pass (log only) | Stats tracking |
| 🟡 **warn** | Don't block; show UI toast | "Amount large, please confirm" |
| 🔴 **block** | Must meet condition to continue; **can override** | "Recipe required" |
| 🔵 **require_approval** | Enter approval workflow | "PO > 100k" |

### 4️⃣ Override Role

"**Who can release blocked actions?**" Common values:

- `manager`
- `admin`
- `null` — no one can override (strictest)

When blocked, UI shows "🔓 Manager Override" button. The role-holder must click + provide reason to release.

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## 📦 Default House Rules (installed automatically)

Out-of-the-box, erpilot ships with 3 common rules:

```
🔴 WO release requires "Recipe"
   trigger: wo.release
   condition: has_bom
   action: block
   override: manager
   message: "Product has no Recipe yet. Please go to Production → Edit Recipe. Or ask manager to
override."

🔵 PO > NT$100k needs manager approval
   trigger: po.create
   condition: amount > 100000
   action: require_approval
   override: manager

🔴 PO must have at least 1 item
   trigger: po.create
   condition: items >= 1
   action: block
   override: none
```

Don't want these? **Just toggle `is_active=false` in UI. No code change, no deploy, no consultant.**

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## 📖 Real Scenario: Buyer's Day

### Scenario: Create a high-value PO

1. **9:00** Buyer opens erpilot "Purchase" page
2. Clicks "➕ Quick Create"
3. Fills: Supplier = ChangJiang / Part = M6 Bolt / Qty = 1000 / Price = 150
4. Saves → System calculates total = NT$150,000
5. **9:01** System auto-runs policy evaluate:
   - "PO > 100k needs manager approval" **hits** (150,000 > 100,000)
   - Action = require_approval
6. Buyer sees:

   > "✅ PO-2026-0042 saved as **draft**, **pending manager approval**"

7. **9:05** Plant Manager receives notification
8. Manager opens "✅ Approvals" → sees this PO in "pending my approval"
9. Clicks "✓ Approve" → adds comment "OK, this supplier has good credit" → confirms
10. PO becomes approved; buyer can proceed with receiving flow

**Throughout the process**:

- ✅ Zero hardcoded
- ✅ Zero consultant fee
- ✅ Zero training
- ✅ Complete audit log (who approved / when / comment)

### Scenario: Emergency override

One day, a customer urgently needs goods, but the product has no Recipe yet:

1. Manager tries to release WO → Blocked
2. System message: "Product has no Recipe. **Or ask manager to override.**"
3. Manager clicks "🔓 Manager Override"
4. Input box: "Please enter override reason (will be audited)"
5. Manager fills: "Customer urgent, release now, will add Recipe tomorrow"
6. Confirm → WO released + audit log records "Manager X overrode rule Y, reason: customer urgent"

**Full compliance, sufficient flexibility**.

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## 🔍 Audit Log (for compliance / debugging)

Every policy evaluation writes to `PolicyAuditLog`:

- Which rule was triggered
- Result (`allowed` / `blocked` / `overridden` / `warned`)
- Context (PO amount / customer ID — **sensitive data auto-truncated**)
- Who triggered / when
- Override user + reason (if any)

For ISO 9001 / GMP / FDA / food safety compliance. Query: `GET /api/policies/audit?rule_id=xxx&trigger=wo.release`

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## ❓ FAQ

### Q1: Can different tenants (companies) have different rules?

✅ Yes. `PolicyRule` includes `tenant_id`; multi-tenant isolation is natural.

## Q2: How are conflicting rules resolved?

For the same trigger, rules are evaluated by **priority ascending**. **First `block` wins**. Convention: `priority < 50` for system rules, `50-200` daily, `200+` special cases.

## Q3: Can I write complex Python conditions?

Don't directly `eval` Python strings (unsafe). Use **`register_condition()` plugin** instead: write a Python function + name + register → it appears in UI.

## Q4: Can I withdraw a rule?

✅ Two ways:

- Set `is_active=false` (recommended; preserves history)
- DELETE removes it (audit log still preserved)

## Q5: Won't managers abuse override?

Every override writes audit log; review periodically. You can add stricter rules:

- E.g., "must email boss before overriding"
- E.g., "max 5 overrides per month per manager"

## Q6: Will rules block system background tasks?

No. erpilot only runs House Rules on **user-initiated** actions. System cron / cleanup / migration don't trigger.

## Q7: I can't read English trigger names (po.create / so.confirm)...

i18n translations coming. Current list has Chinese descriptions. When using AI to author rules, **just talk in plain language**; AI translates trigger names.

## Q8: Can I export/import rules (company move / multi-plant deploy)?

Yes via `GET /api/policies/rules` returning JSON; `POST` to load into a new env. Full import/export UI is Phase 2.

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# 🛠 For Engineers: Technical Details

## Schema

```
class PolicyRule(Base, TenantMixin):
    id: str (UUID PK)
    name: str (max 200)
    description: text
    trigger: str (whitelisted, 16+)
    condition_type: str (5 built-in + plugins)
    condition_params: JSON
    action: str (allow/warn/block/require_approval)
    message: str
    override_role: str | None
    is_active: bool (default True)
    priority: int (default 100)
    created_by / created_at / updated_at

class PolicyAuditLog(Base, TenantMixin):
    id, rule_id, trigger, action_taken
    context: JSON (auto-truncated)
    user_id, override_by, override_reason
    created_at
```

## Integration in service layer

```
# Before (v3.24, hardcoded)
if not bom:
    raise BusinessRuleError("Need BOM")

# After (v3.25, data-driven)
result = await evaluate_policies(db, "wo.release", {"product_id": ...})
if result.blocked:
    raise BusinessRuleError(
        result.message,
        can_override=result.can_override,
        override_role=result.override_role,
    )
```

### Plugin new condition

```python
from app.services.policy_engine import register_condition

async def my_custom(params: dict, context: dict, db: AsyncSession) -> bool:
    # Return True = condition holds, rule passes
    # Return False = condition fails, action triggers
    return ...

register_condition("my_custom", my_custom)
```

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## 🎯 Why this is strategically important for erpilot

> **"Hardcoded rule = customer must edit code = it's not SaaS"**

Previously erpilot hardcoded "WO release needs BOM" — as rigid as SAP/鼎新. From v3.25 onward, with **PolicyEngine**:

- ✅ Users toggle rules in UI → **0 deploy / 0 consultant fee**
- ✅ AI authors rules → **even beginners can customize**
- ✅ Plugin mechanism → **special countries / industries extensible**
- ✅ Audit log → **ISO/GMP/FDA compliant**
- ✅ Manager override → **flexibility + paper trail**

**This is erpilot's real competitive advantage vs SAP/鼎新** — something they can't do, we do: **end users customize their own business rules**.

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## 📚 Related Docs

- USER_MANUAL_EN.md — Full user manual
- HOW_TO_GET_LLM_API_KEY_EN.md — Enable AI chat
- Chinese version: HOUSE_RULES_GUIDE_ZH.md

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*v3.25 (2026-05-18) · erpilot original design · World's first AI-conversational ERP rule engine*